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Sanitizing Door Handle

Abstract

Various implementations of the invention are directed toward a sanitizing door handle and/or components of such sanitizing door handle that sanitizes the door handle during use. Various implementations of the invention are directed toward a gravity-fed sanitizing door handle and/or components of such sanitizing door handle. Various implementations of the invention are directed to a hand sanitizing dispenser that dispenses a fluid, such as a sanitizing fluid, onto hand(s) of a user when the user operates or otherwise engages a door handle, such as, but not limited to, the sanitizing door handles discussed above. In some implementations of the invention, the fluid is dispensed directly onto the hands of the user. In some implementations of the invention, the fluid is dispensed through dispensers and onto the hands of the user through contact with the dispensers. In some implementations of the invention, the fluid is dispensed onto the exterior of the sanitizing door handle and is subsequently passed onto the hands of the user through contact with the sanitizing door handle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This Application is a continuation of U.S. application Ser. No. 17/514,402 entitled “Sanitizing Door Handle,” filed on Oct. 29, 2021. U.S. application Ser. No. 17/514,402 claims priority to U.S. Provisional Patent Application No. 63/107,148, entitled “Sanitizing Door Handle,” filed on Oct. 29, 2020; and U.S. application Ser. No. 17/514,402 claims priority to U.S. Provisional Patent Application No. 63/254,094, entitled “Sanitizing Door Handle,” filed on Oct. 9, 2021. Each of the foregoing patent applications is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION
[0002] The invention is generally related to door handles, and more specifically to sanitizing door handles.
BACKGROUND OF THE INVENTION
[0003] Conventional door handles suffer from actual or perceived cleanliness or sanitation issues. Such door handles may include, but are not limited to, restroom door handles, medical/hospital door handles, refrigerator or other food storage door handles, hotel/motel door handles, airplane bathroom handles, school door handles, clean room door handles, and other door handles.
[0004] In some instances, door handles with improved cleanliness and/or sanitation are needed. In some instances, door handles with fluid dispensers, such as, but not limited to sanitizing fluid dispensers, are needed.
SUMMARY OF THE INVENTION
[0005] Various implementations of the invention are directed toward a sanitizing door handle and/or components of such sanitizing door handle, such that the sanitizing door handle is sanitized or cleaned in some fashion. Various implementations of the invention are directed toward a manual sanitizing door handle and/or components of such sanitizing door handle. In various implementations of the invention, the fluid selected for use with various implementations of the invention controls, in part, the level of “cleaning” or “sanitizing” of the door handle from sterilized to disinfected to cleaned or to some other level of cleanliness.
[0006] Various implementations of the invention are directed to a hand sanitizing dispenser that dispenses a fluid, such as a sanitizing fluid, onto hand(s) of a user when the user operates or otherwise engages a door handle, such as, but not limited to, the sanitizing door handles discussed above. In some implementations of the invention, the fluid is dispensed directly onto the hands of the user. In some implementations of the invention, the fluid is dispensed onto the exterior of the sanitizing handle and is subsequently passed onto the hands of the user through contact with the sanitizing door handle. In various implementations of the invention, the fluid selected for use with various implementations of the invention controls, in part, the level of “cleaning” or “sanitizing” of the hands of the user from sterilized to disinfected to cleaned or to some other level of cleanliness.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1**A and **1**B illustrate a sanitizing door handle from a first perspective and a second perspective, respectively, according to various implementations of the invention.

[0008] FIG. **2** illustrates a dispenser unit that may be used in various implementations of the invention.

[0009] FIG. **3** illustrates a plurality of dispensers implemented in a sanitizing door handle according to various implementations of the invention.

[0010] FIG. **4** illustrates a plurality of dispensers implemented in a sanitizing door handle according to various implementations of the invention.

[0011] FIG. **5** illustrates a plurality of dispensers disposed in a distribution manifold, which in turn is disposed in a sanitizing door handle according to various implementations of the invention.

[0012] FIG. **6** illustrates a plurality of dispensers implemented in a sanitizing door handle according to various implementations of the invention.

[0013] FIG. **7**A and **7**B illustrate a plurality of dispensers implemented on a sanitizing door handle from a first perspective and a second perspective, respectively, according to various implementations of the invention.

[0014] FIG. **8** illustrates a plurality of dispensers implemented in a sanitizing door handle according to various implementations of the invention.

[0015] FIG. **9** illustrates a distribution manifold for a sanitizing door handle according to various implementations of the invention.

[0016] FIG. **10** illustrates a sanitizing door handle according to various implementations of the invention.

[0017] FIG. **11** illustrates a replaceable refill unit according to various implementations of the invention.

[0018] FIG. **12** illustrates an interior of a sanitizing door handle according to various implementations of the invention.

[0019] FIG. **13** illustrates a plurality of dispensers under a single permeable membrane implemented in a sanitizing door handle according to various implementations of the invention.

[0020] FIG. **14** depicts a sanitizing door handle configured for parallel use with an existing door handle according to various implementations of the invention.

[0021] FIG. **15** depicts a sanitizing door handle adapter configured to replace an existing door handle according to various implementations of the invention.

[0022] FIG. **16** depicts a sanitizing door handle according to various implementations of the invention.

[0023] FIG. **17** depicts a sanitizing door handle according to various implementations of the invention.

[0024] FIG. **18** depicts a replaceable refill unit configured to use with a sanitizing door handle according to various implementations of the invention.

[0025] FIGS. **19**A and **19**B depict sanitizing door handles according to various implementations of the invention.

[0026] FIG. **20** depicts a sanitizing door handle cassette according to various implementation of the invention.

[0027] FIG. **21**A depicts a sanitizing door handle cassette installed on a door handle according to various implementations of the invention, and FIGS. **21**B-**21**C depict different adaptors for installing a sanitizing door handle cassette according to various implementations of the invention.

[0028] FIG. **22** depicts a sanitizing door handle according to various implementations of the invention.

[0029] FIG. **23** depicts a distribution manifold partially installed in a door handle housing according to various implementations of the invention.

[0030] FIG. **24** depicts a portion of a distribution manifold according to various implementations of the invention.
[0031] FIG. **25** depicts dispensing applicators with perforated plastic liquid permeable membranes according to various implementations of the invention.
[0032] FIG. **26** illustrates a dispenser unit that may be used in various implementations of the invention.

DETAILED DESCRIPTION

[0033] Various implementations of the invention are directed toward a sanitizing door handle and/or components of such sanitizing door handle such that the sanitizing door handle is sanitized or cleaned in some fashion. Various implementations of the invention are directed toward a manual sanitizing door handle and/or components of such sanitizing door handle. In such implementations, a user manually operates door handle that dispenses a fluid, such as, but not limited to, a sanitizing fluid, onto a grip of the sanitizing door handle during operation of the door handle. In various implementations of the invention, the fluid selected for use with various implementations of the invention controls, in part, the level of “cleaning” or “sanitizing” of the door handle from sterilized to disinfected to cleaned or to some other level of cleanliness.
[0034] Various implementations of the invention are directed to a hand sanitizing dispenser that dispenses a fluid, such as a sanitizing fluid, onto hand(s) of a user when the user operates or otherwise engages a door handle, such as, but not limited to, the sanitizing door handles discussed above. In some implementations of the invention, the fluid is dispensed directly onto the hands of the user. In some implementations of the invention, the fluid is dispensed onto the exterior of the sanitizing handle and is subsequently passed onto the hands of the user through contact with the sanitizing door handle. In various implementations of the invention, the fluid selected for use with various implementations of the invention controls, in part, the level of “cleaning” or “sanitizing” of the hands of the user from sterilized to disinfected to cleaned or to some other level of cleanliness.
[0035] Various implementations of the invention are directed to a gravity-fed hand sanitizing dispenser that dispenses a fluid, such as a sanitizing fluid, onto a door handle and/or hand(s) of a user when the user operates or otherwise engages the door handle, such as, but not limited to, the sanitizing door handles discussed above.
[0036] In some implementations of the invention, the fluid may be a sanitizing fluid that includes any fluid having sterilizing, disinfecting, antiseptic, antimicrobial, and/or other sanitizing or cleaning properties. Active ingredients in the sanitizing fluid may include alcohol-based sanitizing fluids such as, but not limited to, isopropyl alcohol, ethyl alcohol, n-proponal, and other alcohol-based fluids; or non-alcohol-based sanitizing fluids such as, but not limited to, benzalkonium chloride (a chemical disinfectant), triclosan (an antiseptic), thymol (an organic antibacterial agent), and/or other non-alcohol-based sanitizing fluids. In some implementations of the invention, the active ingredients may be mixed with a carrier ingredient such as glycerol or other gel-like ingredient to achieve a certain amount of viscosity as would be appreciated. In some implementations of the invention, the active ingredients may be aerosolized to be sprayed onto hand(s) or door handles as would be appreciated. Sanitizing fluids are available from various manufacturers and sold under names, including, but not limited to: Purell®, Germ-X®, CleanWell™, Babyganics®, CVS®, BlumNaturals®, and other manufacturers and/or names.
[0037] FIG. **1**A illustrates a sanitizing door handle **100** from a first perspective and FIG. **1**B illustrates sanitizing door handle **100** from a second perspective, according to various implementations of the invention. In some implementations of the invention, door handle **100** is a gravity-fed sanitizing door handle, though in other implementations, door handle **100** may employ a pump. Door handle **100** includes a door handle housing **110**, a fluid container **120**, and a fluid container coupler **115**. In some implementations of the invention, fluid container **120** may include a fluid container neck **125**. In some implementations of the invention, door handle housing **110** may be fabricated from a variety of materials, including, but not limited to: metal, plastic, ceramic,

fiberglass, wood, polymer, rubber, or other materials that may be used to fabricate door handle housing **110**, and/or combinations thereof, as would be appreciated. In some implementations of the invention, door handle housing **110** forms a substantially hollow interior housing chamber (not illustrated in FIG. **1**). In some implementations, door handle housing **110** may be formed from one or more door housing components. In some implementations, door handle housing **110** may also be formed to house fluid container **120**. In some implementations of the invention, at least one door housing component is movable and/or detachable from door handle housing **110** in order to provide access to interior housing chamber. In some implementations of the invention, door handle housing **110** is detachable from a door in order to provide access to interior housing chamber. Interior housing chamber and its purpose is described in further detail below.

[0038] Door handle **100** may be configured as a grip, a knob, a lever, a button, a pushbar, a pullbar, a latch, or other suitable handle by which a user opens, closes, latches, unlatches, locks, unlocks, activates, or otherwise operates door handle **100** to open or close a door to which door handle **100** is operatively attached. In some implementations of the invention, door handle **100** may be configured to push open the door. In some implementations of the invention, door handle **100** may be configured to pull open the door. In some implementations of the invention, door handle **100** may be configured to push to actuate a latch to open the door. In some implementations of the invention, door handle **100** may be configured to pull to actuate the latch to open the door. In some implementations of the invention, door handle **100** may be configured to rotate up to open the door. In some implementations of the invention, door handle **100** may be configured to rotate down to open the door. In some implementations of the invention, door handle **100** may be configured to rotate up to actuate the latch to open the door. In some implementations of the invention, door handle **100** may be configured to rotate down to actuate the latch to open the door. In some implementations of the invention, door handle **100** may be configured to slide left to open the door. In some implementations of the invention, door handle **100** may be configured to slide right to open the door. In some implementations of the invention, door handle **100** may be configured to slide left to actuate the latch to open the door. In some implementations of the invention, door handle **100** may be configured to slide right to actuate the latch to open the door. In some implementations of the invention, door handle **100** may be configured to be depressed to open the door. In some implementations of the invention, door handle **100** may be configured to be depressed to actuate the latch to open the door. In some implementations of the invention, door handle **100** may be configured to open the door or actuate the latch in other manners as would be appreciated.

[0039] According to various implementations of the invention, door handle **100** includes a number of sanitizing liquid applicating dispensers **130** (illustrated as a dispenser **130**A, a dispenser **130**B, a dispenser **130**C, . . . and a dispenser **130**N). In some implementations, dispensers **130** are formed into door handle housing **110**; in some implementations, dispensers **130** are individual standalone sub-components that snap-fit, or otherwise couple into door handle housing **110**; in some implementations, dispensers **130** are disposed in a manifold (not otherwise illustrated in FIG. **1**) that in turn snap-fit, slide-fit, or otherwise couple onto or into door handle housing **110**. According to various implementations of the invention, dispensers **130** comprise a liquid permeable, or semi-permeable membrane that when compressed (i.e., pressured applied), releases a flow of the fluid to and through the membrane, and subsequently, into contact with door handle housing **110** and/or a user's hand.

[0040] FIG. **2** illustrates a standalone dispenser unit **200** (i.e., individual standalone sub-component liquid applicating dispenser **130** referenced above) that may be used in various implementations of the invention. The description of various aspects of dispenser unit **200** may apply to other dispensers **130** that may be used in other implementations as would be appreciated. In some implementations of the invention, dispenser unit **200** includes a liquid permeable membrane **210**. In some implementations, the membrane may be a porous material. In some implementations, the membrane may be a cloth-like material. In some implementations, the membrane may be a sponge-like material. In some implementations, the membrane may be a non-porous, or semi-porous, material with perforations. In some implementations, the membrane may be a plastic material with perforations.

[0041] In some implementations of the invention, liquid permeable membrane **210** may include an outer liquid permeable membrane and an inner liquid permeable membrane (not separately illustrated). In some implementations, the outer membrane may be a porous material. In some implementations, the outer membrane may be a cloth-like material. In some implementations, the outer membrane may be a sponge-like material. In some implementations, the outer membrane may be a non-porous, or semi-porous, material with perforations. In some implementations, the outer membrane may be a plastic material with perforations. In some implementations, the inner membrane may be a porous material. In some implementations, the inner membrane may be a cloth-like material. In some implementations, the inner membrane may be a sponge-like material. In some implementations, the outer membrane may be protective material and the inner membrane may be an absorbent material. In some implementations, the outer membrane may be cloth-like material and the inner membrane may be a sponge-like material. In some implementations, the inner membrane and/or the outer membrane may be other permeable membrane materials as would be appreciated. Various other combinations of the inner and outer membranes described may be used as would be appreciated.

[0042] In some implementations of the invention, dispenser unit **200** includes a dispenser housing **230** and a dispenser spring **240** disposed inside dispenser housing **230**. In some implementations of the invention, dispenser spring **240** includes a spring head **246** with a spring interface **244**. In some implementations of the invention, dispenser housing **230** includes an output opening **232**, an input opening **236**, a spring seat **234**, and a spring retainer **237**. In some implementations of the invention, fluid is received into an interior of dispenser housing **230** via input opening **236**. In some implementations of the invention, fluid is dispensed out of dispenser housing **230** via output opening **232**. In some implementations of the invention, spring head **246** prevents fluid from being dispensed out of output opening **232** of dispenser housing **230**; in this closed “default” position, spring **240** resides in a compressed state between spring retainer **237** and spring seat **234**. In some implementations of the invention, spring interface **244** presses against spring seat **234** to prevent fluid from dispensing out of output opening **232**. In some implementations of the invention, when spring **240** is compressed (e.g., by a user pressing spring head **246** through membrane **210**), spring interface **244** disengages from spring seat **234** to allow fluid to dispense out of output opening **232**.

[0043] While the figures illustrate dispenser **130** and dispenser unit **200** as having a circular shape, various implementations of the invention may use other shapes and/or sizes (or multiple different shapes and/or sizes), for example, depending upon the nature, shape and/or size of door handle **100**, or other factors, as would be appreciated. Also, while the figures illustrate multiple dispensers **130** or multiple dispenser units **200**, various implementations may use more or fewer numbers of such dispensers/dispenser units, including a single dispenser/dispenser unit, for example, depending upon the nature, shape and/or size of door handle **100**, or other factors, as would be appreciated.

[0044] In some implementations of the invention, dispenser unit **200** may be an off-the-self product commercially marketed and sold as a “Dab-O-Matic Applicator” from Dabomatic Corporation, Mount Vernon, New York, www.dabomatic.com.

[0045] FIG. **26** illustrates a standalone dispenser unit **2600** (i.e., individual standalone sub-component liquid applicating dispenser **130** referenced above) that may be used in various implementations of the invention. The description of various aspects of dispenser unit **2600** may apply to other dispensers **130** that may be used in other implementations as would be appreciated. In some implementations of the invention, dispenser unit **2600** includes a push ball **2610** and a dispenser spring **2620** disposed inside a dispenser housing **2630**. In some implementations of the invention, dispenser spring **2620** includes a spring head **2626** with a spring interface **2624**. In some implementations of the invention, dispenser housing **2620** includes a spring output opening **2622**, a push ball output opening **2623**, an input opening **2628**, a spring seat **2625**, and a spring retainer **2627**. In some implementations of the invention, fluid is received into an interior of dispenser housing **2620** via input opening **2628**. In some implementations of the invention, fluid is dispensed out of dispenser housing **2620** through push ball output opening **2623** via spring output opening **2622**.

[0046] In some implementations of the invention, spring head **2626** prevents fluid from being dispensed out of spring output opening **2622** of dispenser housing **2620** and push ball **2610** prevents fluid from being dispensed out of push ball output opening **2623**; in this closed "default" position, spring **2620** resides in a compressed state between spring retainer **2627** and spring seat **2625**. In some implementations of the invention, spring interface **2624** presses against spring seat **2625** to prevent fluid from dispensing out of spring output opening **2622**. In some implementations of the invention, spring **2620** presses push ball **2610** against a push ball retainer **2633** to prevent fluid from dispensing out of push ball output opening **2623**. In some implementations of the invention, when push ball **2610** is pressed, thereby compressing spring **2620** (e.g., by a user pressing push ball **2610**), spring interface **2624** disengages from spring seat **2625** and/or push ball **2610** disengages from push ball retainer **2633** to allow fluid to dispense out of spring output opening **2622** and/or push ball output opening **2623**.

[0047] In some implementations of the invention, dispenser unit **2600** may include an optional membrane or other covering over push ball **2610**, similar to those described above with regard to membrane **210** as would be appreciated (not otherwise illustrated in FIG. **26**).

[0048] In some implementations of the invention, dispenser unit **2600** may also be an off-the-self product commercially marketed and sold by Dabomatic Corporation. Other dispenser units from this and other companies may be similarly utilized and/or configured for use with various implementations of the invention as would be appreciated. Various implementations of the invention are described with reference to dispenser unit **200**; however, dispenser unit **2600**, or any other dispenser, may be used in place of or conjunction with dispenser unit **200** as would be appreciated.

[0049] While the figures illustrate dispenser **130** as having a circular shape, various implementations of the invention may use other shapes and/or sizes (or multiple different shapes and/or sizes), for example, depending upon the nature, shape and/or size of door handle **100**, or other factors, as would be appreciated. Also, while the figures illustrate multiple dispensers **130**, various implementations may use more or fewer numbers of such dispensers/dispenser units, including a single dispenser/dispenser unit, for example, depending upon the nature, shape and/or size of door handle **100**, or other factors, as would be appreciated.

[0050] FIG. **3** illustrates a plurality of dispensers **130**, each comprising a dispenser unit **200** (illustrated as a dispenser unit **200**A, a dispenser unit **200**B, a dispenser unit **200**C, and a dispenser unit **200**N) in door handle **100** according to various implementations of the invention. In some implementations of the invention, door housing **110** of door handle **100** forms an interior chamber **310** which receives fluid from fluid container **120** (of FIG. **1**). In some implementations of the invention, each dispenser unit **200** receives fluid from interior chamber **310** through, for example, input opening **236** (illustrated in FIG. **2**) and subsequently dispenses fluid through output opening **232** (illustrated in FIG. **2**) when spring **240** is further compressed.

[0051] FIG. **4** illustrates a plurality of dispensers **130** implemented in door handle **100** according to various implementations of the invention. In such implementations, door handle housing **110** may be configured to replace and/or provide various aspects and/or functionality of dispenser housing **230** of FIG. **2**. In particular, a portion of door handle housing **110** (e.g., a left wall of door handle housing **110** in FIG. **4**) provides the functionality of spring retainer **237** and output opening **232** and spring seat **234** may be formed from and into door handle housing **110** to provide similar operation of springs **240** (illustrated as a spring **240**A, a spring **240**B, a spring **240**C, . . . and a spring **240**N) as would be appreciated. In some implementations of the invention, door handle housing **110** of door handle **100** forms interior chamber **310** which receives fluid from fluid container **120** and subsequently dispenses fluid through an output opening formed in door handle housing **110** when dispensers **130** are engaged by a user and spring(s) **240** are further compressed. As would be appreciated, door hand housing **110** may be similarly configured to replace and/or provide various aspects and/or functionality of dispenser housing **2620** of FIG. **26**.

[0052] FIG. **5** illustrates a plurality of dispensers **130**, each implemented as a dispenser unit **200**, disposed in a distribution manifold **500**, which in turn is disposed in door handle **100** according to various implementations of the invention. More particularly, distribution manifold **500** is disposed in interior chamber **310** formed by door handle housing **110** of door handle **100** according to various implementations of the invention. In some implementations of the invention, distribution manifold **500** is removably and/or replaceably disposed in interior chamber **310**. In some implementations of the invention, distribution manifold **500** includes a manifold wall **510** which forms an interior manifold chamber **520**. In some implementations of the invention, interior manifold chamber **520** receives fluid from fluid container **120** as would be appreciated and dispenses fluid from manifold chamber **520** through input opening(s) **235** and output opening(s) **232** of dispense unit(s) **200** as described above.

[0053] FIG. **6** illustrates a plurality of dispensers **130**, each implemented as spring **240** disposed in distribution manifold **500**, which in turn is disposed in interior chamber **310** formed by door handle housing **110** of door handle **100** according to various implementations of the invention. In some implementations of the invention, distribution manifold **500** is removably and/or replaceably disposed in interior chamber **310**. In some implementations of the invention, distribution manifold **500** includes manifold wall **510** which forms interior manifold chamber **520**. In some implementations of the invention, interior manifold chamber **520** receives fluid from fluid container **120** as would be appreciated and dispenses the fluid from manifold chamber **520** through dispensers **130** as described above with regard to FIG. **4**.

[0054] FIG. **7**A illustrates a manifold sleeve **700** according to a first perspective; and FIG. **7**B illustrates manifold sleeve **700** according to a second perspective, each according to various implementations of the invention. In some implementations of the invention, manifold sleeve **700** attaches to an exterior of door handle **100** (as illustrated in FIG. **7**B). Manifold sleeve **700** may snap fit, slide fit, couple, affix, adhere, velco, or otherwise secure to an exterior of door handle **100** in a variety of ways as would be appreciated. In various implementations of the invention, manifold sleeve **700** provides various features of the invention without significant modifications to door handle **100**. Such implementations may include manifold sleeve **700**, fluid container **120**, and a flexible coupler (not illustrated in FIG. **7**) between the two, where fluid container **120** is attached to the door, typically near door handle **100**.

[0055] Such implementations accommodate a distribution manifold on an exterior of door handle **100**, rather than in an interior of door handle **100** as would be appreciated.

[0056] FIG. **8** illustrates a plurality of dispensers **130** disposed in door handle **100** in accordance with various implementations of the invention. In some implementations of the invention, an optional interior wall **830** separates door handle housing **110** into two portions: a first handle portion **810** and a second handle portion **820**. Interior wall **830** restricts fluid to an interior of first handle portion **810** and not to an interior of second handle portion **820**, thereby restricting fluid to the portion of door handle **100** in which dispensers **130** are disposed. In such implementations, a user engaging first handle portion **810** would contact dispenser(s) **130** and thus, fluid, whereas a user engaging second handle portion **820** would not contact dispenser(s) **130** and may not necessarily contact fluid as would be appreciated.

[0057] FIG. **9** illustrates a distribution manifold **900** according to various implementations of the invention. In some implementations of the invention, distribution manifold **900** comprises a flexible manifold wall portion **910**A and a substantially rigid manifold wall portion **910**B. As would be appreciated, substantially rigid manifold wall portion **910**B is configured to accommodate dispensers **130** (or dispenser units **200**) in output openings **232**.

[0058] FIG. **10** illustrates a door handle **1000** according to various implementations of the invention. In some implementations of the invention, door handle **1000** includes a fluid container **120**, a door latch mechanism **1030**, and a door handle housing **1010**. In some implementations, fluid container **120** stores a fluid, such as, but not limited to, a sanitizing fluid, as discussed above, and feeds a distribution manifold and associated dispensers (not otherwise illustrated in FIG. **10**) housed in door handle housing **1010**. As illustrated, door handle housing **1010** includes two portions: a first handle portion **1010**A and a second handle portion **1010**B. According to various implementations, first handle portion **1010**A houses the distribution manifold and associated dispensers (not otherwise illustrated in FIG. **10**), whereas, second handle portion **1010**B does not. The distribution manifold in door handle housing **1010** restricts fluid to first handle portion **1010**A and not to second handle portion **1010**B, thereby restricting fluid to the portion of door handle **100** in which dispensers **130** are disposed. In such implementations, a user engaging first handle portion **1010**A would contact dispenser(s) **130** and thus, fluid, whereas a user engaging second handle portion **1010**B would not contact dispenser(s) **130** and may not necessarily contact fluid as would be appreciated.

[0059] FIG. **11** illustrates a replaceable refill unit **1100** according to various implementations of the invention. According to various implementations, replaceable refill unit **1100** comprises a fluid container **120** with fluid, a coupler **1130**, and a distribution manifold **500** with dispensers **130** (as illustrated as dashed lines due to being obscured by distribution manifold **500**). According to various implementations, replaceable refill unit **1100** consists essentially of fluid container **120** with fluid, coupler **1130**, and distribution manifold **500** with dispensers **130**. According to various implementations of the invention, replaceable refill unit **1100** may be replaced in/on door handle **100** from time to time, as fluid is dispensed and used in connection with door handle **1000**. In some implementations of the invention, replaceable refill unit **1100** consists essentially of fluid container **120** with fluid and distribution manifold **500** with dispensers **130**; where coupler **1130** is configured for reuse as would be appreciated. In some implementations of the invention, replaceable refill unit **1100** consists essentially of fluid container **120** with fluid and a plurality of dispensers **130**, including, but not limited to dispenser unit **200**; where coupler **1130** and distribution manifold **500** are configured for reuse as would be appreciated. In some implementations of the invention, replaceable refill unit **1100** consists essentially of distribution manifold **500** with dispensers **130**; where coupler **1130** is configured for reuse and where fluid container **120** is configured for refilling (as opposed to replacing) as would be appreciated. In some implementations of the invention, replaceable refill unit **1100** consists essentially of dispensers **130** or dispenser units **200**; where distribution manifold **500** and coupler **1130** are configured for reuse and where fluid container **120** is configured for refilling (as opposed to replacing) as would be appreciated.

[0060] While fluid container **120** and distribution manifold **500** are depicted in a perpendicular arrangement (fluid container **120** in a vertical orientation and distribution manifold **500** in a horizontal orientation), fluid container **120** and distribution manifold **500** may be arranged in any other configuration depending on type of door handle **100** or application as would be appreciated. As would also be appreciated, coupler **1130** may be configured differently to accommodate other such configurations as would be appreciated. In some implementations, coupler **1130** may be formed of flexible tubing to accommodate installation with a variety of door handles **100**; in some implementations, coupler **1130** may be formed from injection molded material to accommodate installation with specific door handle(s) **100**; or in some implementations, coupler **1130** may be formed from other materials or other forming processes as would be appreciated.

[0061] FIG. **12** illustrates an interior of sanitizing door handle **1000** according to various implementations of the invention. In some implementations of the invention, door handle housing

1210 (illustrated in FIG. **12** as a housing portion **1210**A and a housing portion **1210**B) may open to reveal the interior of sanitizing door handle **1000** and to provide access to replaceable refill unit **1100**. In some implementations of the invention, once door handle housing **1210** is open, replaceable refill unit **1100** (including fluid container **120**, coupler **1130**, and/or distribution manifold **500** with dispensers **130**) may be replaced with a new replaceable refill unit **1100**, or various components thereof, as would be appreciated. In some implementations of the invention, door handle housing **1210** may open for purposes of installing sanitizing door handle **1000** onto an existing door latch mechanism as would be appreciated.

[0062] FIG. **13** illustrates a plurality of dispensers **130**, each implemented as spring **240** disposed in distribution manifold **500**, which in turn is disposed in interior chamber **310** formed by door handle housing **110** of door handle **100** similar to that of FIG. **6**. However as illustrated in FIG. **13**, springs **240** are configured under a single permeable membrane to dispense the fluid from manifold chamber **520** according to various implementations of the invention.

[0063] FIG. **14** depicts a sanitizing door handle configured for parallel use with an existing door handle according to various implementations of the invention. As depicted in FIG. **14**, sanitizing door handle **100** may be configured to mount to an existing door handle **1400** by one or more brackets **1410** (illustrated as a bracket **1410**A and a bracket **1410**B). In this configuration, a user may use either sanitizing door handle **100** or existing door handle **1400** to open the door. In some implementations of the invention, bracket(s) **1410** may be adjustable or vary in size to fit a range of existing door handles **1400**. While sanitizing door handle **100** is depicted as being mounted on a right side of existing door handle, sanitizing door handle **100** may be configured to mount on a left side, or either side, of existing door handle according to some implementations of the invention. Similar implementations may apply to existing horizontal door handles as would be appreciated.

[0064] FIG. **15** depicts a sanitizing door handle adapter configured to replace an existing door handle according to various implementations of the invention. As depicted in FIG. **15**, sanitizing door handle **100** may be configured to mount to a spindle **1500** of an existing latch mechanism (not otherwise illustrated) of a door, thereby replacing an existing door handle. In some implementations, sanitizing door handle **100** may be mounted to spindle **1500** by an adaptor **1510** as would be appreciated. In some implementations, adaptor **1510** may include a spring to assist with return rotation of sanitizing door handle **100** after operation as would be appreciated. In some implementations of the invention, adaptor **1510** may be adjustable or vary in size to fit a range of existing door handles. Sanitizing door handle **100** may be configured to replace either, or both, left-opening door handles or right-opening door handles as would be appreciated.

[0065] FIG. **16** depicts a sanitizing door handle according to various implementations of the invention. As depicted in FIG. **16**, sanitizing door handle **100** replaces an existing door handle in its entirety according to various implementations of the invention. In some implementations, sanitizing door handle **100** replaces an existing door handle by mounting to an existing spindle and latch mechanism such as described above with reference to FIG. **15**. In some implementations, sanitizing door handle **100** may be used in place of a conventional door handle. In some implementations, sanitizing door handle **100** may be configured as either, or both, a left-opening door handle or a right-opening door handle as would be appreciated.

[0066] FIG. **17** depicts a sanitizing door handle according to various implementations of the invention. As depicted in FIG. **17**, sanitizing door handle **100** is configured as a door handle pull according to various implementations of the invention. Sanitizing door handle **100** comprises fluid container **120** and door handle housing **110** including a plurality of dispensers **130**. In some implementations of the invention, sanitizing door handle **100** includes an optional door plate **1710** and/or an optional drip tray **1720**. As depicted, in some implementations, door handle housing **110** may include a first portion **1730** in which dispensers **130** are disposed, and a second portion **1740** without any dispensers. In such implementations, users may selectively grasp (or engage) either first portion **1730** (and subsequently contact dispensers **130** and the fluid) or second portion **1740** as would be appreciated. As depicted, dispensers **130** may be disposed facing the door, though other orientations of dispensers may be used as would be appreciated.

[0067] FIG. **18** depicts a replaceable refill unit configured to use with a sanitizing door handle according to various implementations of the invention. In FIG. **18**, replaceable refill unit **1100** is depicted as displaced from door handle **100** to illustrate its removal/installation vis-à-vis door handle **100**. As depicted, replaceable refill unit **1100** comprises fluid container **120**, coupler **1130**, and distribution manifold **500** including a plurality of dispensers **130** disposed therein according to various implementations of the invention. In some implementations, replaceable refill unit **1100** may include fewer or additional components as described above. In some implementations, door handle **100** is configured to removably receive replaceable refill unit **1100** within door handle housing **110**. In some implementations, when in place in door handle **100**, replaceable refill unit **1100** resides within door handle housing **110**. In some implementations, when in place in door handle **100**, at least distribution manifold **500** resides within door handle housing **110**. In some implementations, door handle housing **110** includes a recess **1810** (corresponding to first portion **1730**) that exposes dispensers **130** when replaceable refill unit **1100** is installed in door handle housing **110**.

[0068] FIGS. **19**A and **19**B depict sanitizing door handles according to various implementations of the invention. More specifically, FIG. **19**A depicts a “clip-on” adaptor **1910**A that attaches to a vertical door handle **100** in accordance with various implementations of the invention. In some implementations of the invention, adaptor **1910**A is configured to receive a sanitizing door handle cassette **1920**A, which in turn is configured to receive distribution manifold **500**, including one or more dispensers **130**. In some implementations of the invention, adaptor **1910**A is configured to receive cassette **1920**A, which is itself distribution manifold **500**. Various configurations of adaptor **1910**A may be used to attach or affix cassette **1920**A to door handle **100** as would be appreciated. FIG. **19**B depicts an adaptor **1910**B that is originally manufactured (or modified) as part of door handle **100** to receive a sanitizing door handle cassette **1920**B in accordance with various implementations of the invention. In some implementations of the invention, adaptor **1910**B is configured to receive cassette **1920**B, which is itself distribution manifold **500**. In some implementations, adaptor **1910**B (or its function) may be formed into and/or part of door handle **100** as would be appreciated.

[0069] FIG. **20** depicts a sanitizing door handle cassette **1920** according to various implementation of the invention. According to various implementations of the invention, cassette **1920** comprises one or more dispenser **130** and/or a fluid container **120**. In some implementations, cassette **1920** may be a single use cassette (i.e., disposable once fluid in fluid container is depleted. In some implementations, cassette **1920** may be refillable by a factory or recycle center as would be appreciated. In some implementations, cassette **1920** may be refillable at a point of use as would be appreciated.

[0070] FIG. **21**A depicts a sanitizing door handle according to various implementations of the invention. More specifically, FIG. **21**A depicts a “clip-on” adaptor **2110** that attaches to horizontal door handle **100** in accordance with various implementations of the invention. In some implementations of the invention, adaptor **2110** is configured to receive sanitizing door handle cassette **1920**, which in turn is configured to receive distribution manifold **500**, including one or more dispensers **130**. In some implementations of the invention, adaptor **2110** is configured to receive cassette **1920**, which is itself distribution manifold **500**. Various configurations of adaptor **2110** may be used to attach or affix cassette **1920** to door handle **100** as would be appreciated.

[0071] FIG. **21**B depicts clip-on adaptor **2110**A without cassette **1920** in accordance with various implementations of the invention. FIG. **21**C depicts an adaptor **2110**B **1920**, where adaptor **2110**B is originally manufactured (or modified) as part of door handle **100** to receive sanitizing door handle cassette **1920** in accordance with various implementations of the invention. In some implementations of the invention, adaptor **2110**B is configured to receive cassette **1920**, which is itself distribution manifold **500**. In some implementations, adaptor **2110**B (or its function) may be formed into and/or part of door handle **100** as would be appreciated.

[0072] FIG. **22** depicts a sanitizing door handle **100** according to various implementations of the invention. Door handle **100** may include a fluid container **120** (only the bottom portion of which is depicted in FIG. **22**). In some implementations of the invention, fluid container **120** is cylindrical in shape to facilitate installation. For example, fluid container **120** may be threaded into door handle housing **110** or into distribution manifold **500** (not illustrated in FIG. **22**) which may be disposed in door handle housing **110** as would be appreciated. As depicted, door handle **100** has an upper portion that dispenses sanitizing fluid in accordance with various implementations of the invention, and a lower portion that does not dispense sanitizing fluid. In some implementations of the invention, distribution manifold **500** occupies the upper portion of door handle housing **110** to facilitate dispensing of sanitizing fluid.

[0073] FIG. **23** depicts a distribution manifold **500** partially installed in a door handle housing **110** according to various implementations of the invention. FIG. **24** depicts a retaining portion **2410** of distribution manifold **500** according to various implementations of the invention. In some implementations of the invention, a screw **2310** is inserted through door handle housing and into retaining portion **2410** of distribution manifold **500** to secure distribution manifold **500** in place. Other mechanisms for retaining distribution manifold **500** in door handle housing **110** may be used, including snap fit, retaining clips, etc., as would be appreciated.

[0074] FIG. **25** depicts dispensers **130** with perforated plastic membranes **2510** according to various implementations of the invention. As depicted, membranes **2510** include one or more perforations which allow fluid to flow through plastic membrane **2510** as would be appreciated. Various numbers of perforations, patterns, sizes, etc., may be used to control fluid flow through plastic membrane **2510** as would be appreciated. In some implementations, membrane **2510** may be comprised of a non-porous or semi-porous material other than plastic, including but not limited to, metal, rubber, ceramic, etc., as would be appreciated; these other non-porous or semi-porous materials may be perforated as would be appreciated.

[0075] While various implementations of the invention have been described herein in reference to certain door handles illustrated in the drawings, for example, other door handles and configurations thereof may be used as would be appreciated.

[0076] While various implementations of the invention have been described herein in reference to door handles and doors, other implementations of the invention may be configured for use with barriers other than doors, such as but not limited to, gates, pass-throughs, security checkpoints, or other barriers, where users regularly contact a particular surface for access through or past the barrier.

Claims

1. A replaceable refill unit comprising: a fluid container configured to retain a sanitizing fluid; and a distribution manifold having at least one dispenser disposed therein, the distribution manifold configured to distribute the sanitizing fluid from the fluid container to the dispenser, the dispenser configured to dispense the sanitizing fluid.

2. The replaceable refill unit of claim 1, further configured to be disposed within or affixed to a door handle.

3. The replaceable refill unit of claim 1, wherein the sanitizing fluid is gravity-fed from the fluid container to the distribution manifold.

4. The replaceable refill unit of claim 1, wherein the fluid container and the distribution manifold are removably coupled to one another.

5. The replaceable refill unit of claim 1, wherein the fluid container and the distribution manifold are affixed to one another.

6. The replaceable refill unit of claim 1, wherein the at least one dispenser comprises a plurality of dispensers.

7. The replaceable refill unit of claim 1, wherein each dispenser comprises a spring configured to prevent flow of the sanitizing fluid out of the dispenser when in a first compressed state and to dispense the sanitizing fluid when further compressed by a user.

8. The replaceable refill unit of claim 1, wherein the fluid container is refillable.

9. The replaceable refill unit of claim 1, wherein the distribution manifold is configured to be disposed within a door handle.

10. The replaceable refill unit of claim 1, wherein the distribution manifold is configured to be disposed within a door handle such that the door handle provides a sanitizing portion and a non-sanitizing portion.

11. The replaceable refill unit of claim 1, wherein each dispenser comprises a liquid permeable membrane through which the sanitizing liquid is dispensed.

12. The replaceable refill unit of claim 1, wherein each dispenser comprises a perforated plastic membrane through which the sanitizing liquid is dispensed.

13. The replaceable refill unit of claim 1, further comprising an adaptor configured to removably affix the replaceable refill unit to an exterior of a door handle.

14. The replaceable refill unit of claim 1, further comprising an adaptor configured to removably affix the replaceable refill unit to an interior of a door handle.

15. An apparatus comprising: a distribution manifold configured to be affixed to or disposed within a door handle and configured to removably attach to a sanitizing fluid container, wherein the distribution manifold comprises at least one dispenser, wherein the distribution manifold is further configured to distribute a sanitizing fluid from the sanitizing fluid container to the at least one dispenser via gravity, wherein the dispenser operatively dispenses the sanitizing fluid to a user.

16. The apparatus of claim 15, wherein the at least one dispenser comprises a spring configured to prevent flow of the sanitizing fluid out of the at least one dispenser when in a first compressed state and to dispense the sanitizing fluid when further operatively compressed by a user.

17. The apparatus of claim 15, wherein the at least one dispenser comprises a liquid permeable membrane through which the sanitizing liquid is dispensed.

18. The apparatus of claim 15, wherein the at least one dispenser comprises a perforated plastic membrane through which the sanitizing liquid is dispensed.